



PhD in Biology, Chemistry, Health (BCS) with experience in computational (molecular dynamics simulations, molecular docking, and high-performance computing (HPC)) and experimental techniques (transmembrane protein production and purification from *E. coli* bacteria). Strong background in Bio-health engineering, pharmaceutical biotechnology and biochemistry. Passionate about computational modeling, theoretical chemistry, drug discovery, and structural biology.

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Education and Diplomas

PhD in Biology, Chemistry, Health (10/2022 – 12/2025)

University of Reims Champagne Ardenne (URCA). Reims, France.

International Master (MSc) in Bio Health engineering (2021–2022)

University of Grenoble Alpes (UGA). Grenoble, France.

National School of Biotechnology (ENSB) (2015-2020)

MSc in Pharmaceutical Biotechnology. Constantine, Algeria.

BEng in Biotechnology. Constantine, Algeria.

Professional experiences

PhD Research – Biology, Chemistry, Health (2022 – 12/2025)

ICMR – CNRS 7312, University of Reims, France

Optimization The protocol of production and purification of transmembrane protein (TSPO) from *E. coli* bacteria.

Quantify and Qualify analysis of TSPO protein using SDS PAGE and UV-Vis spectrophotometry.

Developed and optimized python scripts to process and analyze high-throughput data from large scale molecule dynamics.

Optimize automated workflow to execute batch for docking and IGMPlot density analysis.

Small molecule structural optimization using Gaussian and ORCA software.

Internship: (Jan 2022 – June 2022)

LRB-Inserm1039 Laboratory Grenoble, France

Worked on synthesis and biological evaluation of flavone analogs.

Performed NMR, mass spectrometry and chromatography analysis.

Environmental control engineer (Jan 2021– June 2021)

HUPP company, Constantine, Algeria

Conducted environmental impact assessments and quality control.

Research intern: Thesis (Jan2020– June 2020)

Higher National school of Biotechnology, Constantine, Algeria

Worked on extraction and Purification from plants (*physalis L*).

Communications

Organization and communication PhDay (English) (2024, 2025).

Oral communication Journées de ROMEO (2024, 2025) (Frensh).

Oral communication 16èmes (English) *Rencontres du Réseau de Chimie Théorique du Grand Est* (July 2025 | Besançon, France).

Appicom 4th plenary meeting (English) (2023)

Projects & Research

PhD Thesis (10/2022– 12/2025): INTERDY: TSPO Bacteria: Interaction& Dynamic studies

Dr. Luminita Duma

MSc Thesis (Jan 2022 – June 2022): Synthesis and biological evaluation of flavone analogs

Pr. Ahcene Boumendjel

MSc Thesis (Jan2020 – June 2020): Extraction of essential oils from Solanaceae family and antidiabetic activities *in vivo*

Dr. Sara Boussaha

Technical skills

Molecular Dynamics & Docking:

GROMACS, AutoDock, Autodock vina, ChimeraX, PyMOL

HPC & Simulations:

ROMEO & CRIANN, SLURM job management

Spectroscopy & Chromatography:

NMR, UV-Vis, HPLC, MS, IR

Molecular & Cell Biology:

Cell culture, Bacteria culture, protein extraction and purification

PCR, qPCR, Western-Blot, ELISA, SDS-PAGE

Programming & Data Analysis:

Python, R scripts, Excel, MATLAB

Scientific Writing & Documentation:

Word, PowerPoint, ChemDraw

Certifications & Training

Innovate with Design Thinking:

Business plan development

Soft Skills Training:

Teamwork & Leadership

Science Club:

Member in SCF and TSPO club

Softwares

ChemDraw, Excel, Word, PowerPoint, Photoshop, Illustrator,

Video Editing

python, topspin

Languages

English: Advanced

French: Advanced

Arabic: Native

Interests & Hobbies

Sports: Gym, Football, Hiking

Drawing & design

UX Researcher & Consultant (Freelance)

User Research & Testing | Remote